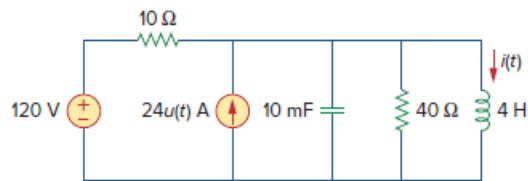


Problem

For the circuit in Fig. 16.67, find $i(t)$ for $t > 0$.

Figure 16.67:



Step-by-step solution

Step 1 of 5

Refer to the circuit in Figure 16.67 in the textbook.

Transform the circuit into s-domain.

$$24u(t) \Rightarrow \frac{24}{s}$$

$$10 \text{ mF} \Rightarrow \frac{1}{sC} = \frac{100}{s}$$

$$4 \text{ H} \Rightarrow sL = 4s$$

Draw the modified circuit as shown in Figure 1.

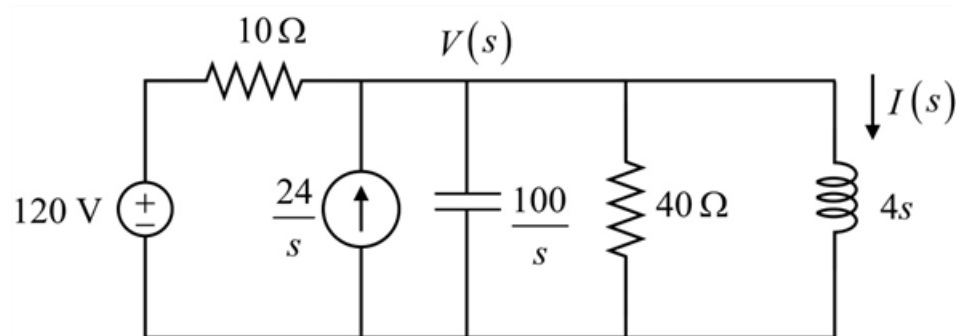


Figure 1

Step 2 of 5

Apply Kirchhoff's current law at the top node.

$$\frac{V(s)-120}{10} + \frac{V(s)}{100/s} + \frac{V(s)}{40} + \frac{V(s)}{4s} = \frac{24}{s}$$

$$V(s) \left[\frac{1}{10} + \frac{s}{100} + \frac{1}{40} + \frac{1}{4s} \right] = \frac{24}{s} + 12$$

$$V(s) \left[\frac{20s + 2s^2 + 5s + 50s}{200s} \right] = \frac{24 + 12s}{s}$$

$$V(s) \left[\frac{2s^2 + 75s}{200} \right] = 24 + 12s$$

$$V(s) \frac{s(2s + 75)}{200} = 12(2 + s)$$

$$V(s) = \frac{2400(s + 2)}{s(2s + 75)}$$

Step 3 of 5

Find the current $I(s)$.

$$I(s) = \frac{V(s)}{4s}$$

Substitute $\frac{2400(s + 2)}{s(2s + 75)}$ for $V(s)$.

$$I(s) = \frac{2400(s + 2)}{4s^2(2s + 75)}$$

$$= \frac{2400(s + 2)}{8s^2(s + 37.5)}$$

$$= \frac{300(s + 2)}{s^2(s + 37.5)}$$

Step 4 of 5

Apply partial fractions.

$$I(s) = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+37.5} \dots\dots (1)$$

$$300(s+2) = As(s+37.5) + B(s+37.5) + Cs^2$$

$$300s + 600 = (As^2 + 37.5sA) + (Bs + 37.5B) + Cs^2$$

Comparing s^0 coefficients,

$$600 = 37.5B$$

$$B = 16$$

Comparing s^1 coefficients,

$$300 = 37.5A + B$$

Substitute 16 for B.

$$300 = 37.5A + 16$$

$$A = 7.573$$

Comparing s^2 coefficients,

$$A + C = 0$$

$$C = -A$$

Substitute 7.573 for A.

$$C = -7.573$$

Step 5 of 5

Substitute 7.573 for A, 16 for B, and -7.573 for C in equation (1).

$$I(s) = \frac{7.573}{s} + \frac{16}{s^2} - \frac{7.573}{s+37.5}$$

Apply inverse Laplace transform.

$$i(t) = (7.573 + 16t - 7.573e^{-37.5t})u(t) \text{ A}$$

Thus, the current is $i(t)$ for $t > 0$ is $\boxed{7.573 + 16t - 7.573e^{-37.5t} \text{ A}}$.